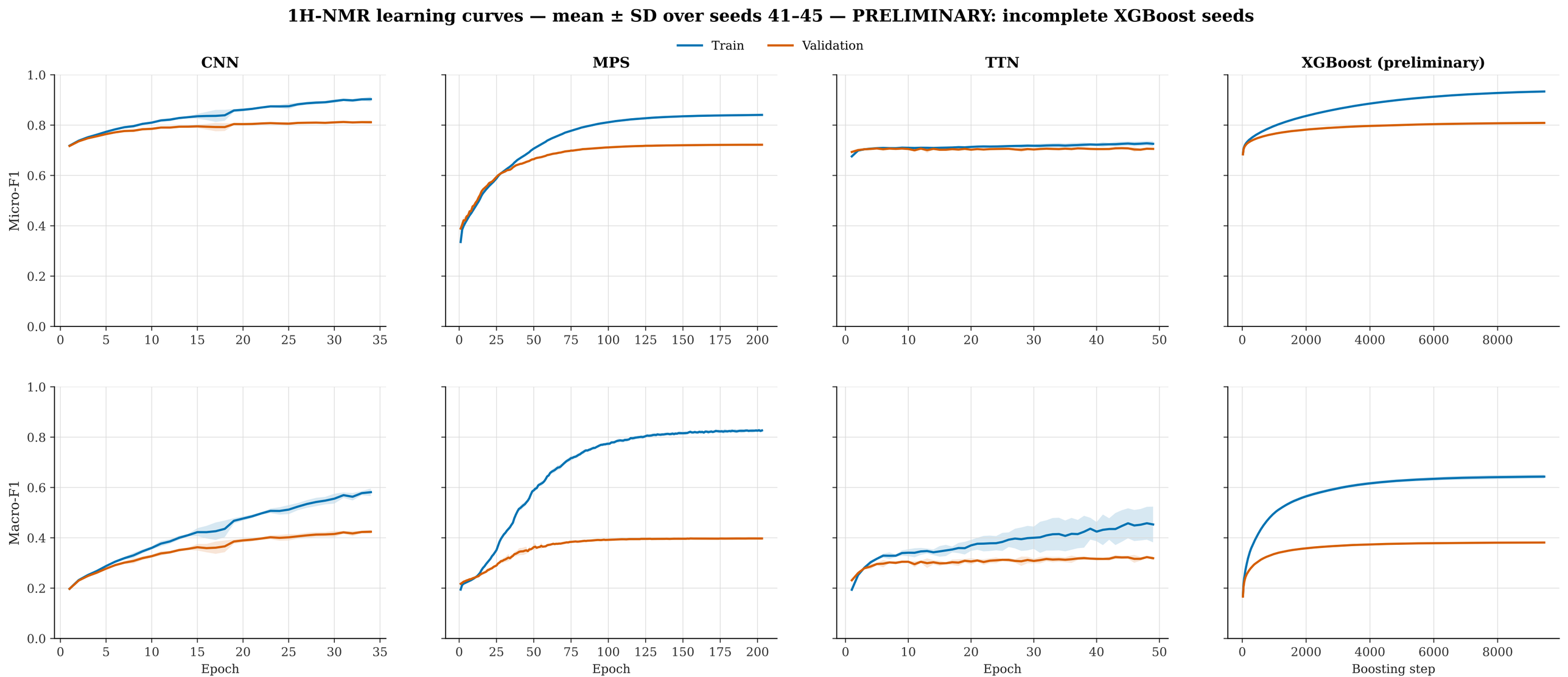

1H-NMR learning curves — mean ± SD over seeds 41–45 — PRELIMINARY: incomplete XGBoost seeds
Train
Validation
CNN
MPS
TTN
XGBoost (preliminary)
Micro-F1
Macro-F1
Epoch
Epoch
Epoch
Boosting step
Native optimization axes; shaded regions show ±1 standard deviation across five seeds.